

Scan for Candidates — Improvement Requirements

Oct 6, 2026 · @Amit

Summary

The scan page currently mislabels its benchmark and tests on a biased universe, so its verdicts can't be trusted yet. Ship SCAN-1 and SCAN-2 first; the rest change what "good" means and how results are shown.

ID	Story	Priority	Why it matters
SCAN-1	Benchmark labeling fix	P0	Cards show the shortfall as if it were holding's return; page contradicts itself
SCAN-2	Point-in-time S&P 500 universe	P0	Using today's members back to 2021 inflates buy-and-hold (survivorship bias)
SCAN-3	Risk-adjusted comparison	P1	Raw return alone always favors holding in a bull market
SCAN-4	Walk-forward testing and regimes	P1	One 70/30 split tests only one market condition
SCAN-5	Concentration transparency	P1	One outlier stock can decide the benchmark
SCAN-6	Robustness across samples	P2	20 stocks × 1.5 years is mostly noise
SCAN-7	Presentation cleanup	P2	Losing or idle templates are ranked as "candidates"

SCAN-1 Benchmark labeling fix (P0)

Story. As a user reading scan results, I want the buy-and-hold benchmark shown as its own return, separate from each template's gap to it, so that I can tell at a glance whether a template won or lost.

Context. In the seed 1471885888 scan, every row satisfies template return – "vs holding" = +58.0%. The column is the gap, but cards read "+22.3% vs –35.7% for holding," which suggests the template won, directly above "Did not beat holding."

Acceptance criteria

1. A benchmark line appears once above the results: "Holding these 20 stocks, equal weight: +58.0% out-of-sample (Apr 2025 – Oct 2026)."
2. Each card reads: "+22.3% after costs vs +58.0% for holding (–35.7 pts)."
3. The table splits the current column into two: "Holding return" (same value every row) and "Difference (pts)".
4. The "Did not beat holding" verdict is computed from the difference, and a unit test asserts it agrees with the sign shown.
5. The benchmark method (equal weight, rebalanced or not, costs applied or not) is stated in the info tooltip.
6. A regression test feeds a known sample and checks template return – difference = holding return for every row.

SCAN-2 Point-in-time S&P 500 universe (P0)

Story. As a user, I want each stock tested only on dates when it was actually in the S&P 500, so that results aren't inflated by picking today's winners.

Context. The sample includes HOOD, which joined the index in 2025, yet is tested from 2021. Today's members are, by construction, the stocks that did well — this biases buy-and-hold upward and every template too.

Acceptance criteria

1. The system stores S&P 500 membership history (ticker, date added, date removed) and refreshes it at least monthly.
2. The random sample is drawn from members as of the scan's start date, including stocks later removed, delisted or acquired.
3. A stock contributes trades and benchmark returns only while it was a member; delisted stocks use their final price (or acquisition price) on exit.
4. Results show "Universe: point-in-time S&P 500 as of 2021-10-08" and list any stocks that left the index during the window.
5. A toggle "Current members only (biased)" remains for comparison, labeled as biased and off by default.

6. A test confirms a known later addition (e.g. HOOD) is excluded before its add date.

SCAN-3 Risk-adjusted comparison (P1)

Story. As a user, I want each template compared with holding on risk as well as return, so that a template that earns less but falls far less isn't marked a failure.

Acceptance criteria

1. For each template and for the holding benchmark, show over the out-of-sample window: total return after costs, annualized return, time in market (%), max drawdown, Sharpe ratio, and return ÷ max drawdown (Calmar).
2. Add an "exposure-adjusted return" column: return per 100% time in market.
3. A template passes the short-term test if it beats holding on return **or** on Sharpe with a max drawdown no deeper than holding's; the card states which test it passed.
4. The holding benchmark row appears in the table with the same columns, pinned at the top.
5. Tooltips define each metric in one plain sentence.
6. Metrics match a reference calculation (e.g. an offline notebook) to within 0.1 pt on a fixed test sample.

SCAN-4 Walk-forward testing and market regimes (P1)

Story. As a user, I want templates tested across several out-of-sample windows and labeled by market regime, so that I can see where a template helps instead of judging it on one bull-market stretch.

Context. The current single 70/30 split puts the 2022 bear market in the fitting period and roughly Apr 2025 – Oct 2026 (mostly rising) in the test period, so templates that sit in cash will almost always lose.

Acceptance criteria

1. Replace the single split with rolling walk-forward windows: fit on 24 months, test on the next 6, step 6 months (configurable).
2. Report per-window results and an aggregate: windows won vs holding, median difference, worst window.

3. Classify each test window as Bull, Bear or Sideways using the S&P 500's return and drawdown in that window (e.g. Bear = index drawdown $\geq 15\%$); show the rule in a tooltip.
4. Add a per-template breakdown: average difference vs holding in Bull, Bear and Sideways windows.
5. Extend the default history to at least 10 years when data allows, so 2018, 2020 and 2022 declines are covered; show the date range used.
6. The long-term test's "window with a major downturn" requirement is checked against the regime labels and states which downturn it used.

SCAN-5 Concentration transparency (P1)

Story. As a user, I want to see how much each stock drove the results, so that I know whether one outlier decided the verdict.

Acceptance criteria

1. A "Stocks in this sample" panel shows, per stock: out-of-sample buy-and-hold return, contribution to the equal-weight benchmark (pts), and trades taken by the top template.
2. Show the median stock's return next to the equal-weight benchmark.
3. Show "Holding without the top contributor" and recompute each template's difference against it.
4. If one stock contributes more than 25% of the benchmark's return, show a warning: "Results depend heavily on [TICKER]."
5. Clicking a ticker opens its price chart with the top template's entries and exits marked.

SCAN-6 Robustness across samples (P2)

Story. As a user, I want a template judged across many random samples, so that one lucky or unlucky draw of 20 stocks doesn't decide its verdict.

Acceptance criteria

1. A "Robustness run" option repeats the scan over N random samples (default 50, max 200) of the chosen size, each with its own recorded seed.
2. Per template, report: % of samples where it beat holding, median difference vs holding, and the 10th–90th percentile range.

3. A "Full universe" option tests every point-in-time S&P 500 member instead of a sample.
4. Users can set sample size (10–100) and filter by sector.
5. Runs longer than 30 seconds execute as a background job with a progress bar and a saved result the user can reopen.
6. The "Chance Sharpe above zero" figure keeps its multiple-testing correction and gains a plain-language label: under 20% "Likely luck," 20–60% "Uncertain," over 60% "Some evidence of edge."

SCAN-7 Results presentation cleanup (P2)

Story. As a user, I want the page to lead with a clear verdict and hide templates that didn't qualify, so that I don't mistake losers for ideas.

Acceptance criteria

1. When no template passes, the top of the results says so in one line, and the short-term and long-term candidate lists collapse under "Show how templates did anyway."
2. Templates below the minimum trade count (30 short-term) are greyed out with the reason "Too few trades (11)."
3. Templates with 0 trades are flagged "Never triggered on this sample" and excluded from rankings.
4. "Later dates" is replaced everywhere by actual dates, e.g. "Out-of-sample: Apr 2025 – Oct 2026."
5. An equity-curve chart shows the top three templates against holding, with drawdowns shaded.
6. Each table column has a tooltip; numbers use consistent signs and units (% for returns, pts for differences).
7. "Open in builder" pre-fills the template and the scan's stocks.

Open questions

- ☐ Which data source provides point-in-time S&P 500 membership and delisted prices (SCAN-2)?
- ☐ Is the holding benchmark rebalanced, and are trading costs applied to it?
- ☐ Walk-forward window lengths: keep 24/6 months, or let users choose?
- ☐ Should the pass rule in SCAN-3 be the same for short-term and long-term templates?

- ☐ Compute budget for robustness runs on EC2 — cap N, or queue jobs?